

LLMOps (Large Language Model Operations)



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Introduction

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What is LLMOps?

LLMOps (Large Language Model Operations) is a set of practices, tools, and workflows used to **deploy, monitor, scale, and continuously improve Large Language Model-based applications in production.**

It ensures that LLMs work **reliably, securely, and cost-effectively** after they move beyond demos and prototypes.



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Why LLMOps ?

Generative AI systems are **dynamic, expensive, and unpredictable**.

LLMOps is critical because it provides the operational discipline needed to **run GenAI applications reliably in real-world production environments**.

Key Challenges in GenAI (Without LLMOps) :

- Inconsistent or hallucinated responses
- High and unpredictable **API/token costs**
- Slow response times under real user load
- No visibility into **what prompts produce bad outputs**
- Security risks (prompt injection, data leakage)

How LLMOps Solves These Problems :

- **Standardizes deployment** of LLM-based services
- Enables **monitoring of latency, cost, output quality**
- Supports **safe updates** to prompts and models
- Adds **guardrails** for security and compliance
- Ensures **scalability** during peak traffic

LLMOps vs MLOps vs DevOps

Aspect	DevOps	MLOps	LLMOps
Primary Focus	Application & infrastructure automation	ML model lifecycle management	Large Language Model-based systems
Main Goal	Fast, reliable software delivery	Deploy & maintain ML models	Run GenAI apps reliably, safely, and cost-effectively
Core Components Managed	Code, CI/CD, servers, containers	Data, training pipelines, models	Prompts, LLM APIs/models, RAG pipelines, tools
Monitoring Metrics	CPU, memory, uptime	Accuracy, drift, latency	Latency, token usage, cost, hallucinations
Cost Considerations	Infrastructure cost	Training & inference cost	Token usage, API cost, retrieval cost
Security Concerns	Infra & API security	Data & model security	Prompt injection, data leakage, access control

LLM Lifecycle: From Prototype to Production

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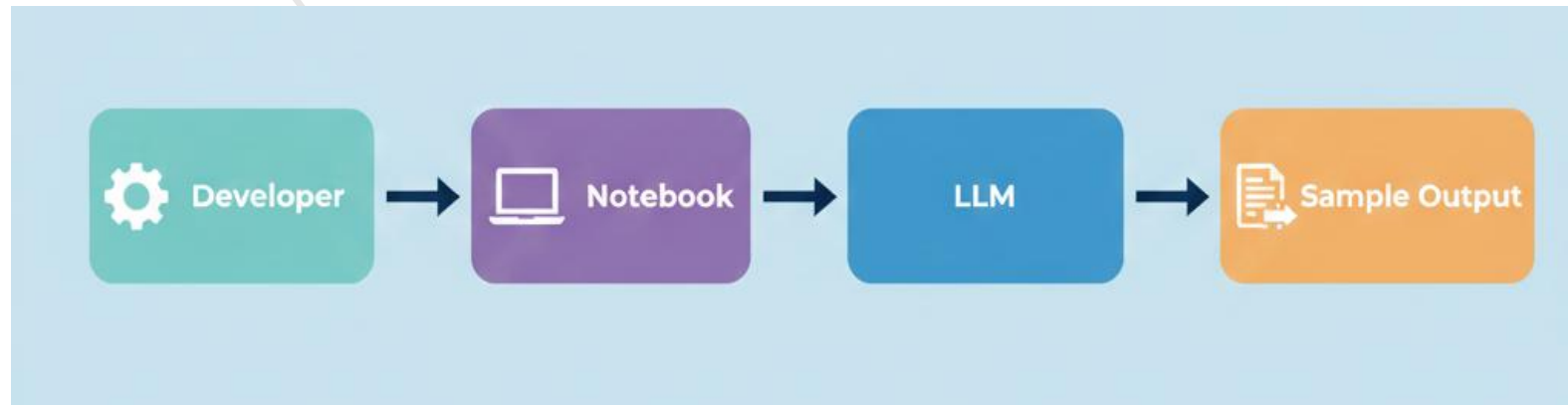
LLM Lifecycle – Prototyping Phase

What is LLM Prototyping?

The **prototyping phase** is where ideas are quickly tested using LLMs to validate **feasibility, usefulness, and response quality** before investing in production systems.

What Happens in This Phase

- Selecting an LLM (OpenAI, Anthropic, open-source)
- Writing basic prompts
- Manual testing via playground or notebooks
- Using sample or small datasets
- No scaling or monitoring yet



Typical Tools Used

- ChatGPT / OpenAI Playground
- Jupyter Notebooks
- LangChain / LlamaIndex (basic usage)
- Local scripts or Postman tests

Limitations of Prototypes

- Works for few users only
- No cost control
- No reliability or security
- Not production-safe

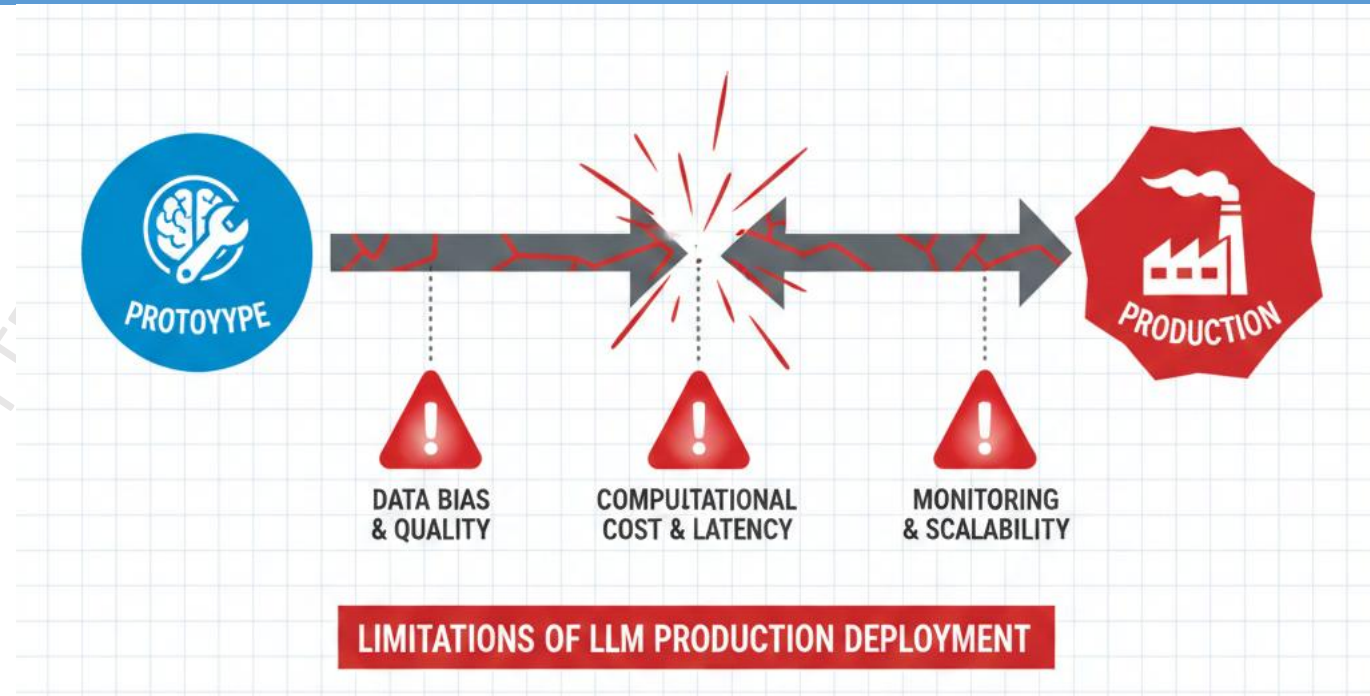
Challenges : Moving LLMs to Production

Why Prototypes Fail in Production

What works in a demo often **breaks** at scale

Key Challenges :

- **Latency:** Slow responses when users increase
- **Cost:** Token usage explodes with real traffic
- **Inconsistency:** Same prompt gives different outputs
- **Hallucinations:** Confident but wrong answers
- **Security risks:** Prompt injection, data exposure
- **Lack of observability:** No idea why outputs fail



Traditional Systems vs LLM Systems :Website: <https://inttrvu.ai>

- Traditional APIs → predictable
- LLM APIs → probabilistic and context-driven

Why LLM Ops Is Required Here :

LLMOps introduces **structure, guardrails, and monitoring** to overcome these challenges.

Production Readiness & Key Design Considerations

Production Readiness Checklist

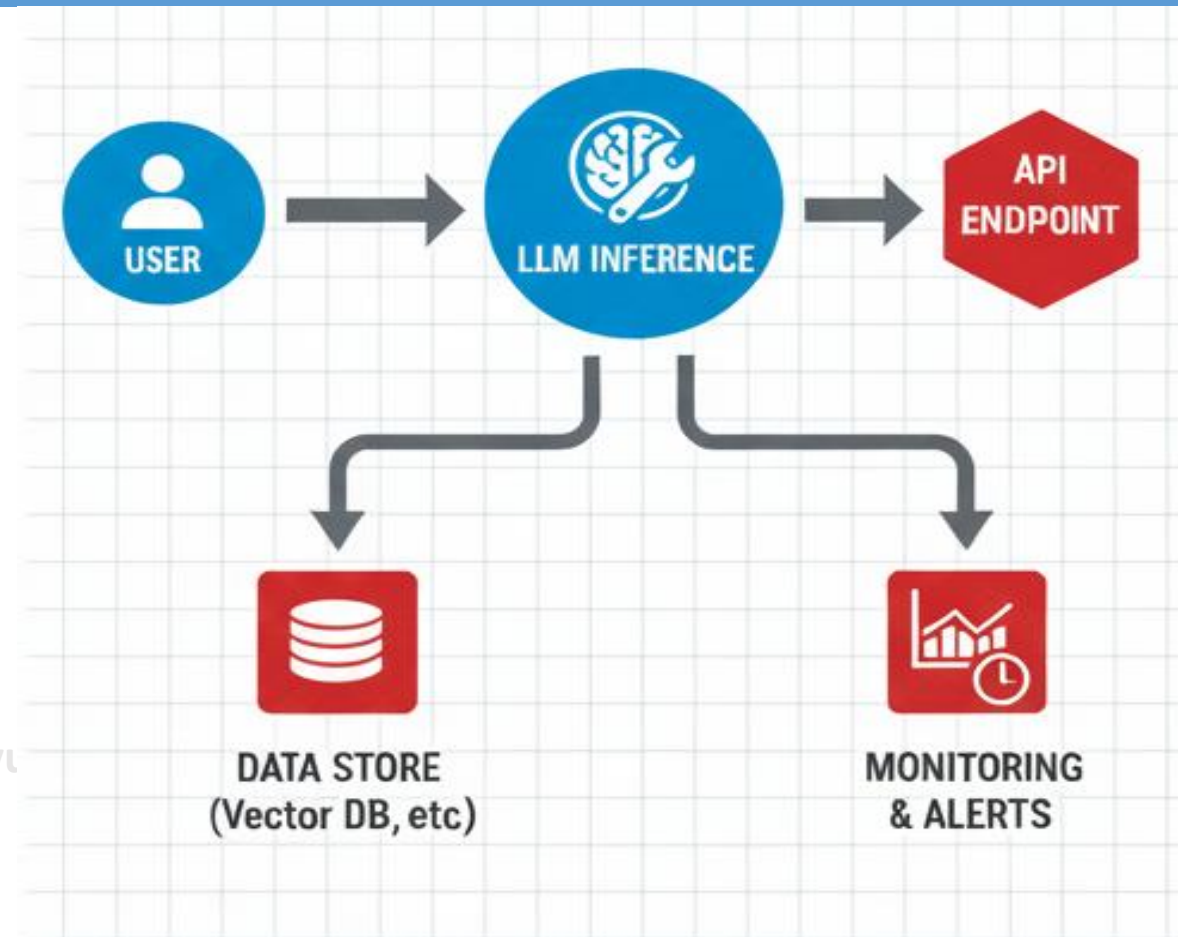
Before going live, ensure:

- API-based serving (not notebooks)
- Prompt versioning
- Cost limits & usage tracking
- Logging of requests & responses
- Error handling & fallbacks
- Secure key management
- Scalable deployment (Docker + Cloud)

Real-World Example

A production RAG chatbot:

- Cached FAQs
- Rate-limited users
- Monitored token usage
- Deployed via Docker on AWS



Key Design Considerations in LLMOps

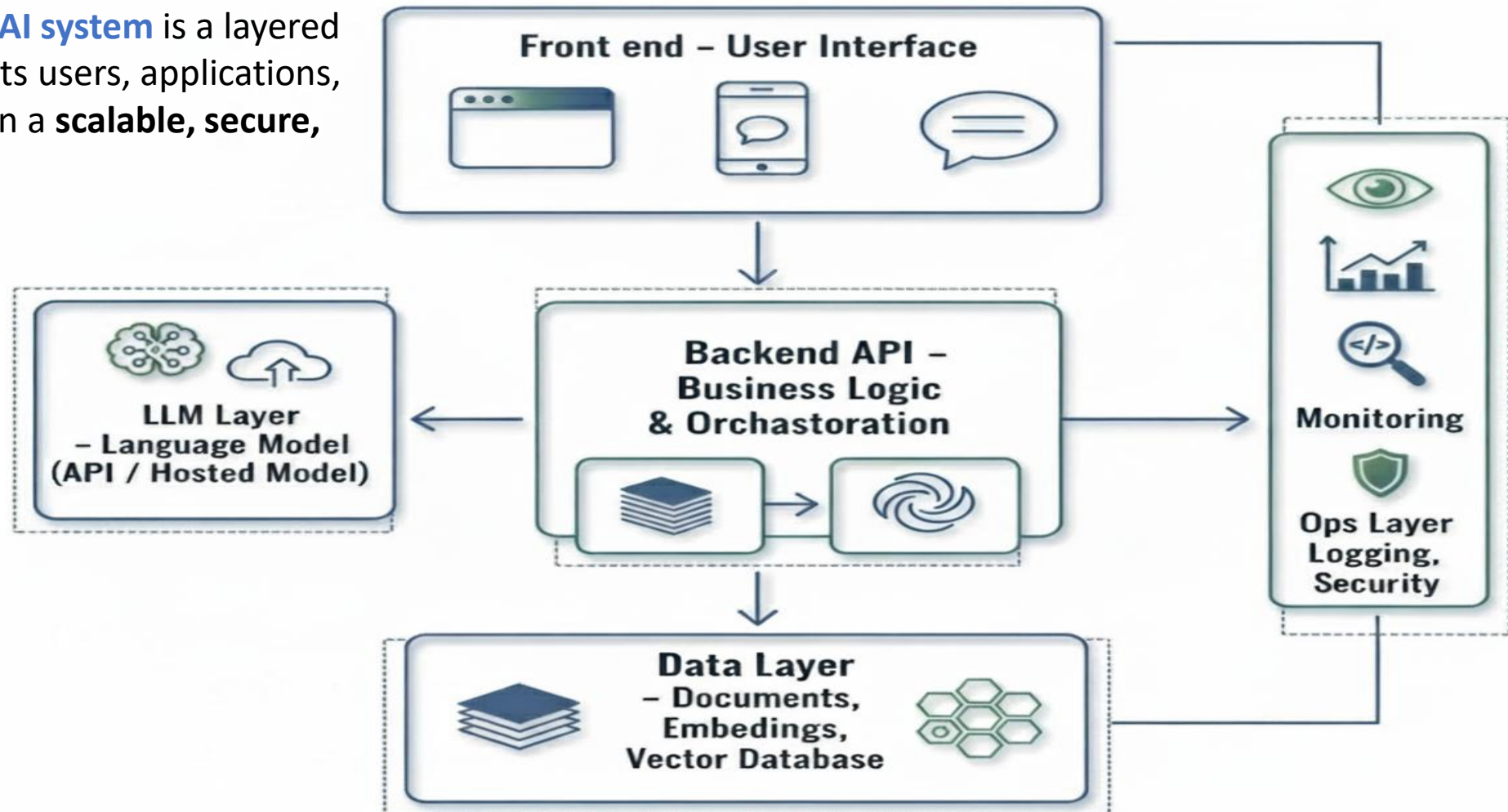
Design Area	What It Means	Techniques / Best Practices	Real-World Example
Latency	Time taken by the LLM to respond to a user request	• Async APIs • Response caching • Smaller context windows	A chatbot uses cached answers for FAQs to respond instantly instead of calling the LLM every time
Cost	Controlling and optimizing token and API usage expenses	• Token budgeting • Prompt optimization • Caching repeated queries	An enterprise limits daily token usage per user to avoid unexpected API bills
Scalability	Ability to handle increasing users and traffic	• Stateless services • Load balancing • Containerized deployments	A GenAI app runs multiple Docker containers behind a load balancer on AWS
Security	Protecting data, prompts, and system access	• Prompt injection protection • Data masking • Role-based access control	An internal chatbot hides sensitive employee data and restricts access by user role

Architecture of Production-Grade GenAI Systems

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High-Level Architecture of a Production-Grade GenAI System

A **production-grade GenAI system** is a layered architecture that connects users, applications, LLMs, and data sources in a **scalable, secure, and observable** way.



Components of Production-Grade GenAI System

Layer	Primary Function	Real-World Example
Frontend	Captures user queries and displays AI responses via UI	A Streamlit web app or chat widget used by employees to ask HR or IT questions
Backend API	Handles business logic, prompt construction, and orchestration	A FastAPI service that receives the question, calls the vector DB, builds the prompt, and sends it to the LLM
LLM Layer	Generates natural language responses	GPT-4 / Claude / LLaMA generating answers to customer support queries
Data Layer	Stores and retrieves documents and embeddings for grounding responses	FAISS / Pinecone storing embeddings of company policies, manuals, and FAQs
Ops Layer	Monitors performance, cost, logs, and enforces security	CloudWatch / Prometheus tracking latency & token usage, IAM controlling access

Data Flow in LLM Applications

Why Data Flow Matters

- Helps debug failures
- Improves performance optimization
- Enables monitoring & auditing

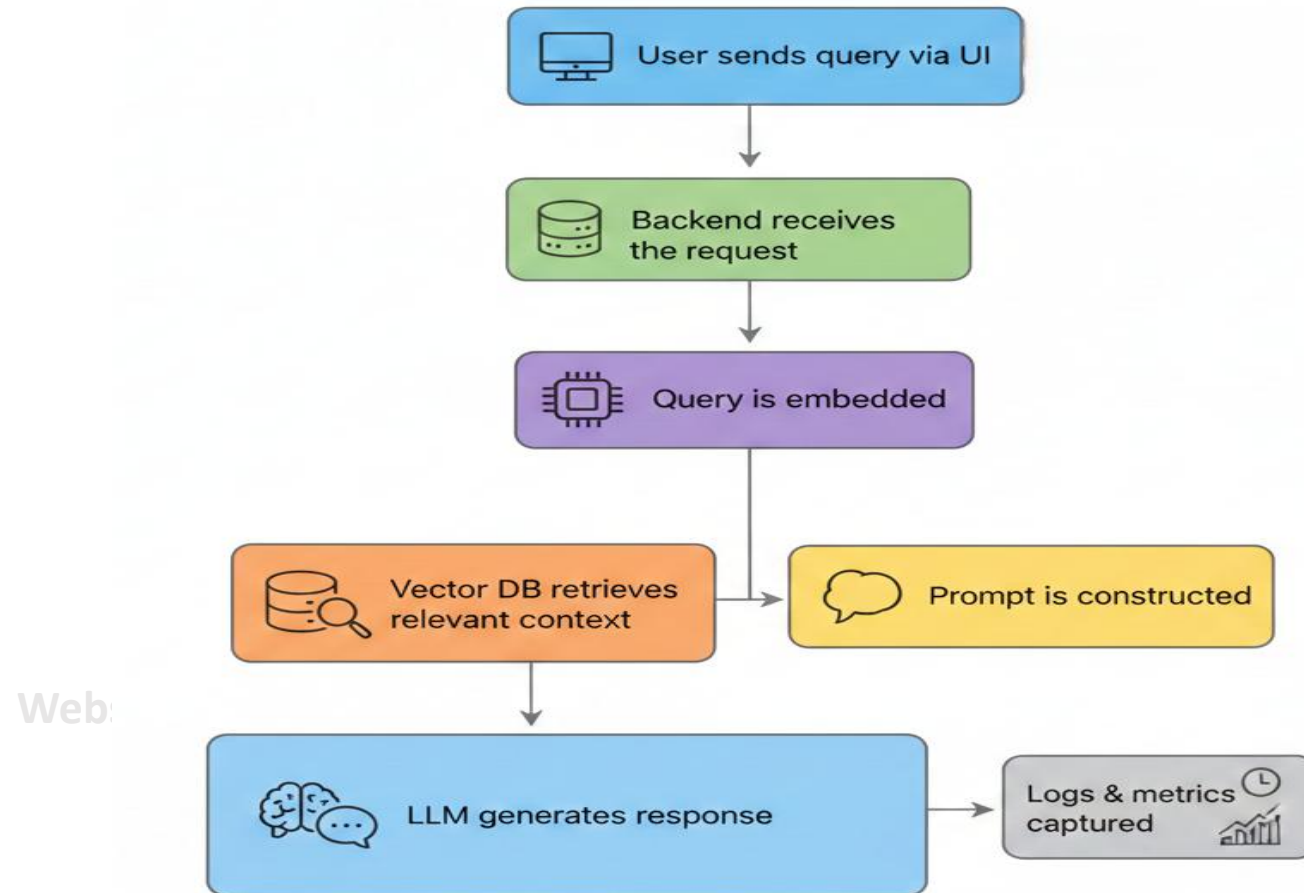
Production Considerations

- Cache frequent queries
- Mask sensitive data
- Log prompts responsibly

Real-World Example

Production chatbot logs:

- Token usage
- Response latency
- Error rates



Building and Serving GenAI Models as Scalable REST APIs

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API-Based LLM Serving

What Is API-Based LLM Serving?

API-based LLM serving means making a Large Language Model available through a **REST API**, so any application can send a request to the model and get a response **easily, consistently, and at scale**.

Why APIs Are Used for LLMs

- Easy integration with web & mobile apps
- Independent scaling of backend and UI
- Cloud-friendly & microservices-ready
- Enables monitoring, security, and versioning



Real-World Example

A customer-support chatbot calls:

POST /generate-response

Backend sends prompt to GPT and returns the answer.

Model Inference Patterns in GenAI Systems

Common Inference Patterns :

- **Synchronous Inference**
The application waits for the LLM to generate a complete response before replying to the user, commonly used in real-time chat applications.
- **Asynchronous Inference**
The request is processed in the background and the result is returned later, suitable for long-running tasks like document summarization or report generation.
- **Streaming Inference**
The LLM sends the response token-by-token as it generates them, creating a faster, ChatGPT-like interactive user experience.
- **Batch Inference**
Multiple prompts are grouped and processed together in a single request to improve throughput and reduce overall inference cost.

Choosing the Right Pattern

- Chat → Streaming / Sync
- Large documents → Async
- High volume → Batch

Real-World Example

Live chat shows answers **word-by-word** using streaming inference.(ChatGPT)

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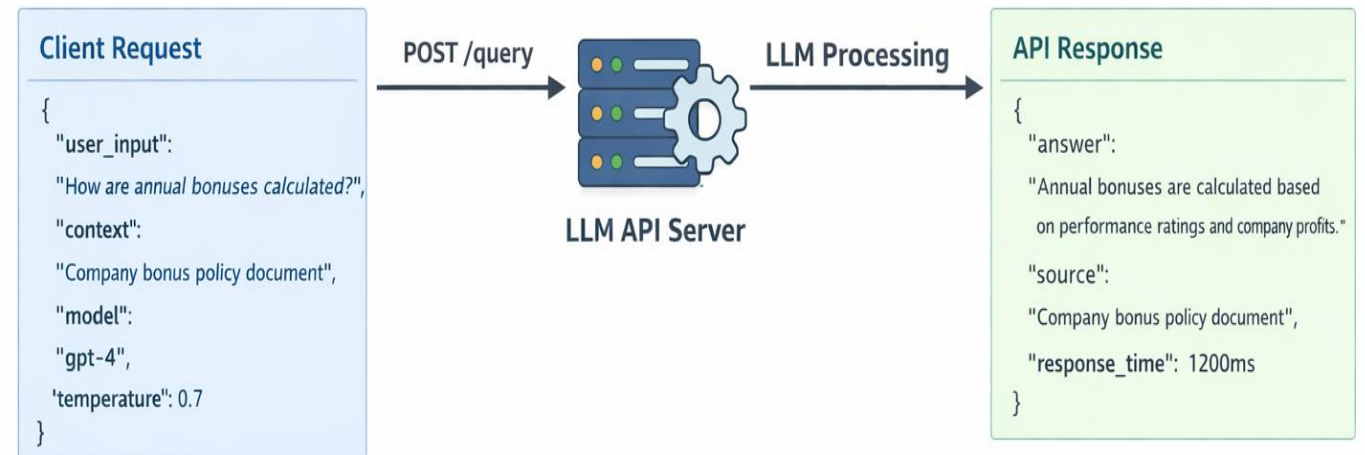
REST API Design for LLM Applications

Key REST Endpoints

- POST /query – user query
- POST /embed – generate embeddings
- GET /health – service health
- POST /feedback – capture user feedback

Best Practices

- Stateless APIs
- Clear request/response schema
- Timeout & retry handling
- Prompt version included in request



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Real-World Example

Backend API accepts:

- User question
- Retrieves context via RAG
- Calls LLM and returns answer

Performance Optimization & Security

Performance Optimization

- **Batching** – process multiple requests together
- **Caching** – reuse frequent responses
- **Async calls** – non-blocking API requests
- **Smaller prompts** – reduce token usage

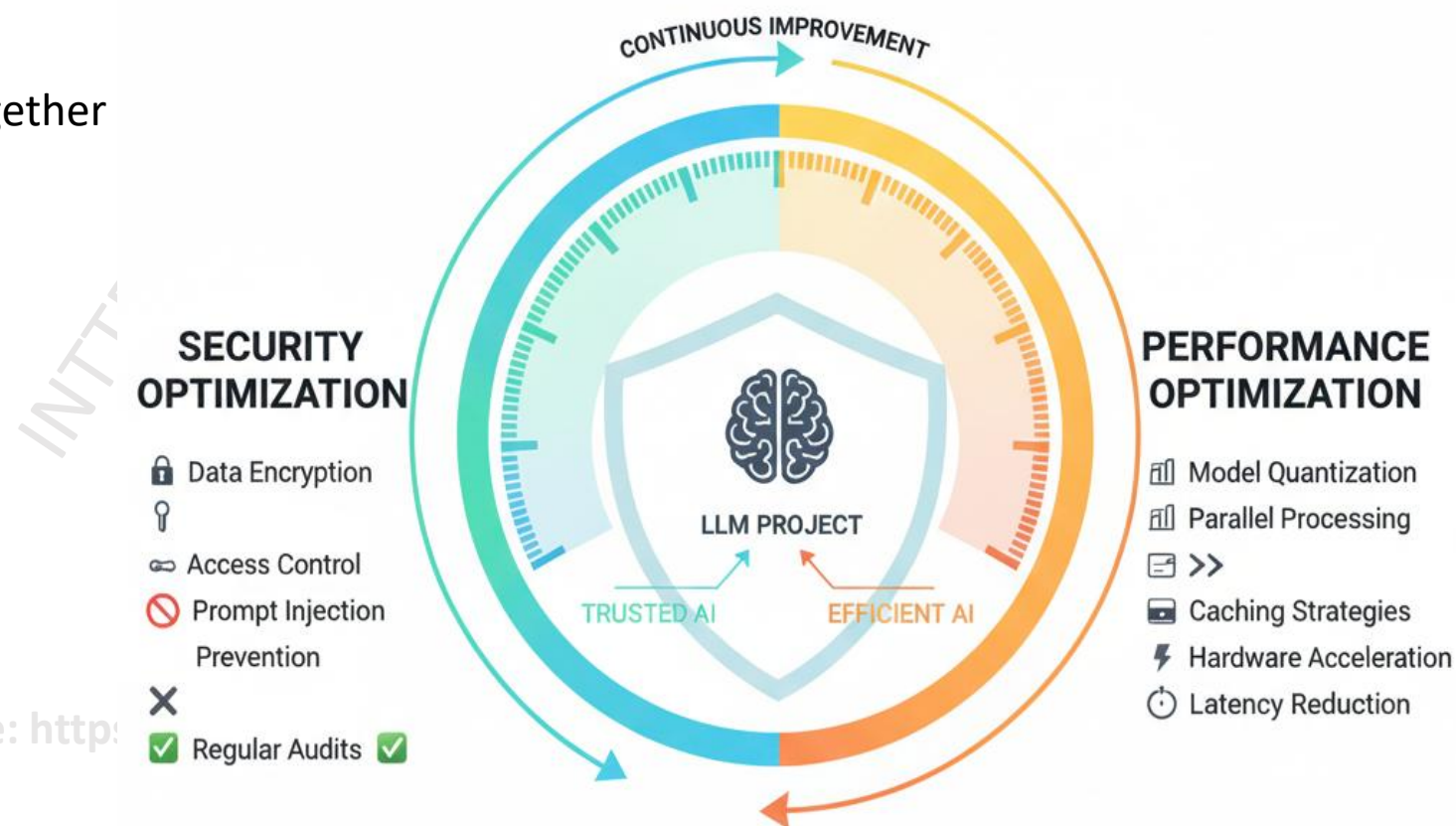
Security & Authentication

- API keys & token-based auth
- Rate limiting per user
- Secure secret storage (env variables)
- Prompt injection protection

Real-World Example

An enterprise API:

- Limits 100 requests/user/min
- Caches FAQ answers
- Masks sensitive data before sending to LLM



Containerizing LLM Applications with Docker

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Why Containerization Is Important in LLMOps ?

What Is Containerization?

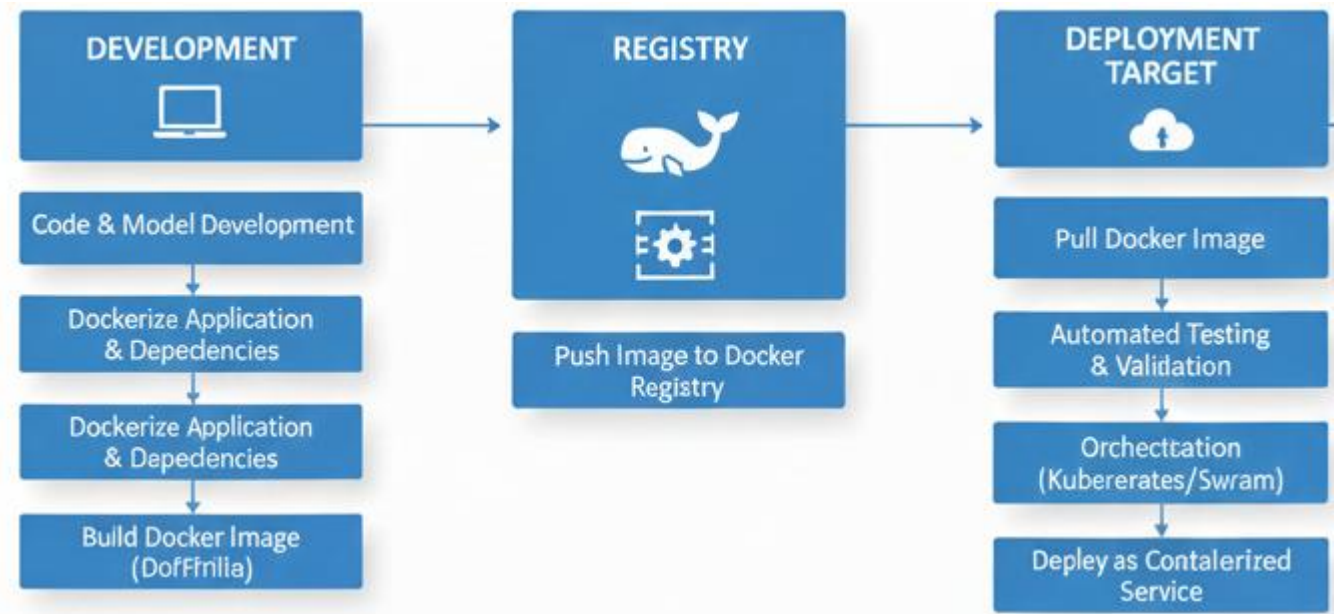
Containerization packages an application **along with its dependencies, libraries, and runtime** into a single portable unit called a **container**.

Why it is required ?

- LLM apps depend on many libraries (LLM SDKs, vector DBs, ML libs)
- Same app must run locally, on cloud, and in CI/CD

Key Benefits

- Consistent behavior across environments
- Easy scaling and deployment
- Faster onboarding for teams
- Cloud & DevOps friendly



Real-World Example

A RAG chatbot runs **identically** on a developer laptop and an AWS EC2 instance using Docker.

Docker Fundamentals for GenAI Applications

Core Docker Concepts :

- **Image** – A read-only blueprint that contains the application code, dependencies, and runtime needed to run the app.
- **Container** – A running instance of an image where the application actually executes.
- **Dockerfile** – A text file that defines step-by-step instructions to build a Docker image.
- **Docker Hub / Registry** – A centralized place to store, share, and pull Docker images.

How Docker Fits in GenAI Apps :

- **Backend API in one container**
Runs the LLM logic, RAG pipeline, and API endpoints independently and can scale without affecting the UI.
- **Frontend (Streamlit) in another container**
Provides the user interface for interacting with the GenAI application without mixing UI and backend code.
- **Vector DB in another container**
Stores and searches embeddings separately, enabling fast retrieval and independent data scaling.

Writing Dockerfiles for LLM Services

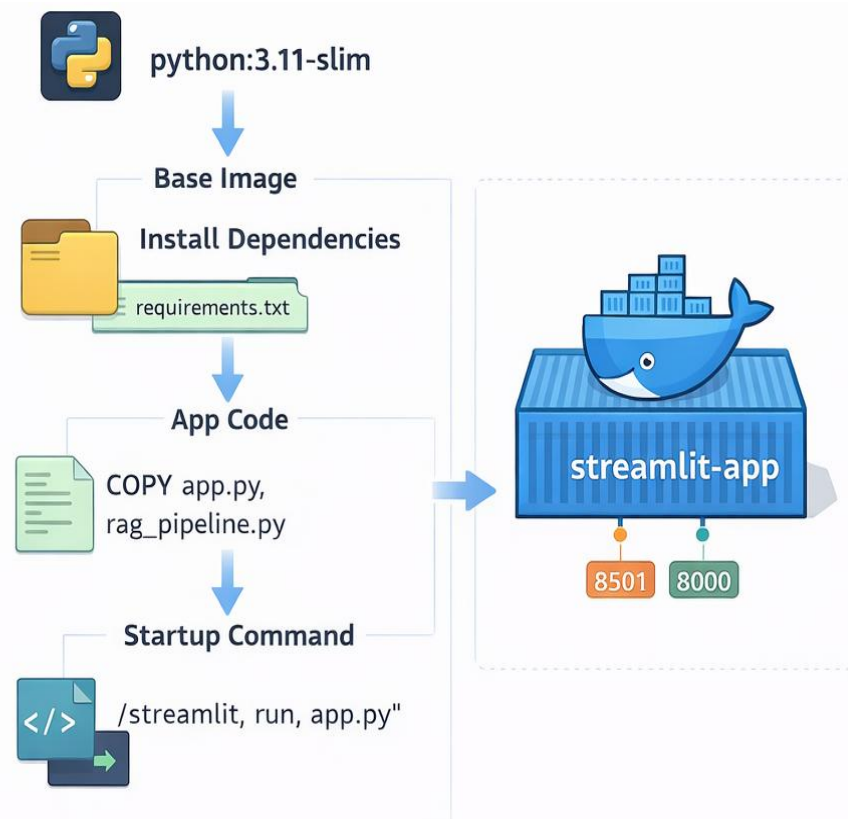
What Is a Dockerfile?

A **Dockerfile** defines how to build a container image step-by-step.

Key Sections in an LLM Dockerfile

- **Base image (Python)** – Defines the runtime environment for the LLM application.
- **Dependency installation** – Installs required libraries and frameworks.
- **Application code** – Copies the GenAI service code into the container.
- **Startup command** – Specifies how the LLM service starts when the container runs.

```
FROM python:3.11-slim
# System deps for PyMuPDF +
# general build tools
RUN apt-get update && apt-get install -y
    build-essential
    python3-dev
    poppler-utils
    libmupdf-dev
    pkg-config && rm -fb /var/lib/apt/lists/*
}
WORKDIR /app
# Install Python deps
COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt
COPY app.py .
COPY rag_pipeline.py .
# Streamlit UI + Prometheus metrics port
EXPOSE 8501
EXPOSE 8000
CMD ["streamlit", "run", "app.py"]
```



Deploying GenAI Services on Cloud Platforms

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Cloud deployment strategies for LLM apps

What Is Cloud Deployment in LLMOps?

Cloud deployment means running GenAI services on cloud infrastructure to achieve **scalability, reliability, and global access**.

Common Deployment Strategies

- **Single VM deployment** – Simple, quick setup (EC2 + Docker)
- **Container-based deployment** – Better scalability (ECS / EKS)

Why Cloud Is Preferred for GenAI ?

- On-demand scaling
- High availability
- Built-in security & monitoring
- Pay-as-you-use pricing

Real-World Example

A startup deploys a RAG chatbot on **AWS EC2 using Docker**, then later migrates to **ECS** (Elastic Container Service) as users grow

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AWS Architecture for GenAI Services

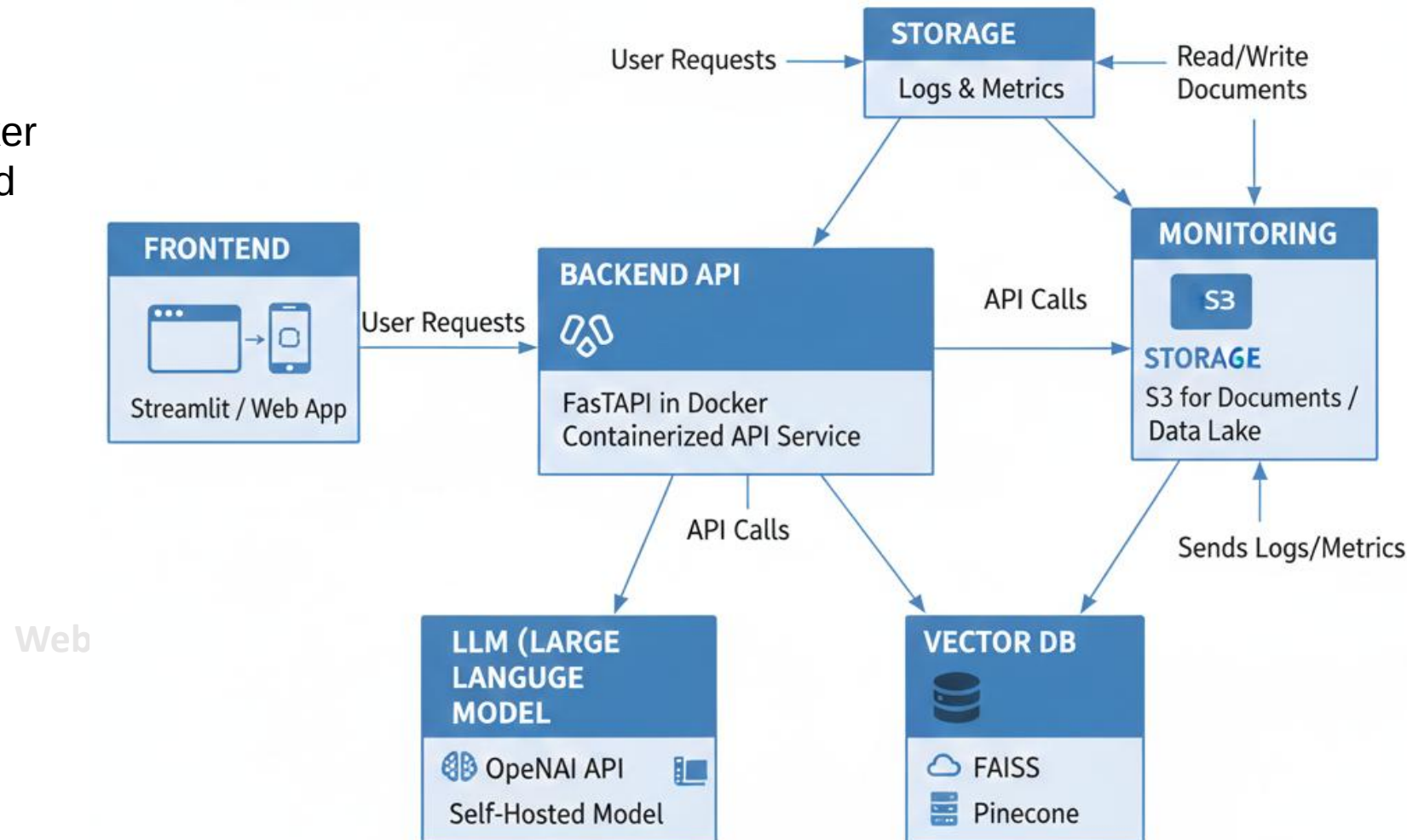
Components

- **Frontend** – Streamlit / Web App
- **Backend API** – FastAPI in Docker
- **LLM** – OpenAI API or self-hosted model
- **Vector DB** – FAISS / Pinecone
- **Storage** – S3 for documents
- **Monitoring** – CloudWatch

Real-World Example

An internal knowledge assistant:

- PDFs stored in S3
- Embeddings in FAISS/Pinecone
- Backend deployed on EC2
- UI exposed via public IP



Compute Options for GenAI Workloads

Aspect	EC2 (Elastic Compute Cloud)	ECS (Elastic Container Service)	EKS (Elastic Kubernetes Service)
What it is	Virtual machines managed by you	AWS-managed container orchestration	Managed Kubernetes service on AWS
Container Support	Manual (Docker setup needed)	Native Docker container support	Kubernetes-native containers
Complexity	Low to Medium	Medium	High
Scalability	Manual or Auto Scaling Groups	Automatic service scaling	Highly scalable with Kubernetes
Best Use Case	Simple deployments, learning, PoC	Production containerized apps	Large-scale, enterprise GenAI platforms
Cost Management	Pay for VM uptime	Pay for running containers	Control-plane + node costs
DevOps Skill Needed	Basic Linux & Docker	Docker + AWS basics	Kubernetes expertise
LLMOps Fit	Good for quick RAG demos	Ideal for production GenAI apps	Best for complex, multi-agents

Monitoring, Logging, and Alerting for Production LLMs

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Why Monitoring Is Critical for LLM Applications

What Is Monitoring in LLM Ops?

Monitoring is the continuous tracking of **performance, cost, quality, and reliability** of LLM-based applications in production.

Why LLM Monitoring Is Different

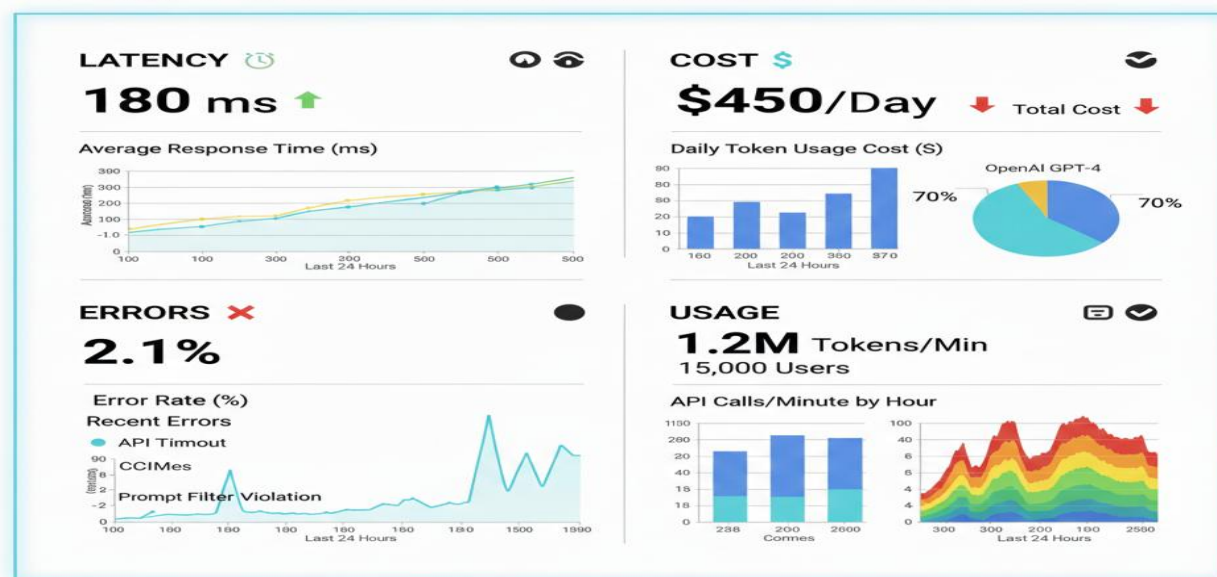
- LLMs are **non-deterministic**
- Costs depend on **token usage**
- Failures may be **silent** (wrong answers, hallucinations)
- Latency can spike unexpectedly

What Can Go Wrong Without Monitoring

- Sudden increase in API bills
- Slow or timed-out responses
- Hallucinated or unsafe outputs
- Undetected system failures



LLM PRODUCTION MONITORING



Metric	What It Tells You
Latency	How fast the LLM responds
Cost	API/token spend per user or request
Errors	Failed API calls, timeouts
Token Usage	Prompt + completion token consumption

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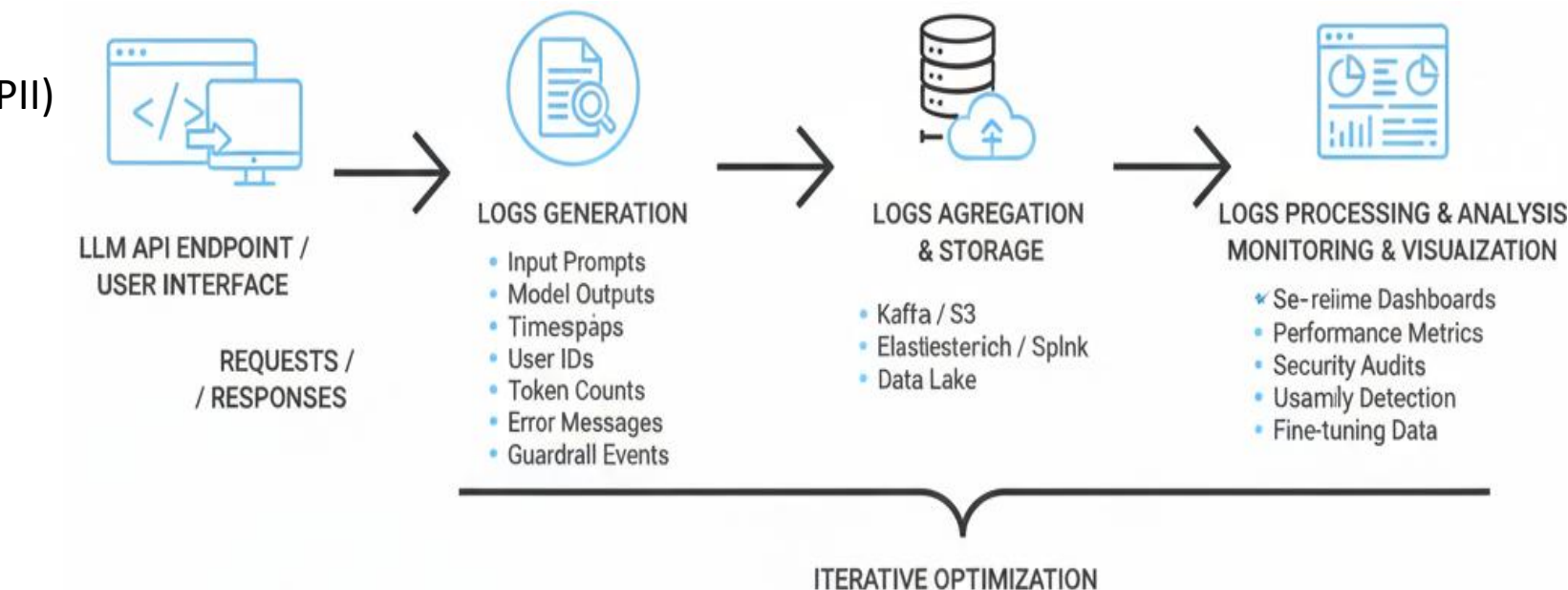
Logging for Production LLMs

What to Log

- User request metadata (not raw PII)
- Prompt version
- Model used
- Token count
- Response status

Why Logging Matters

- Debug wrong answers
- Analyze prompt performance
- Audit system behavior
- Improve quality over time



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Real-World Example

Logs show that a **specific prompt version** causes longer responses and higher cost → prompt optimized.

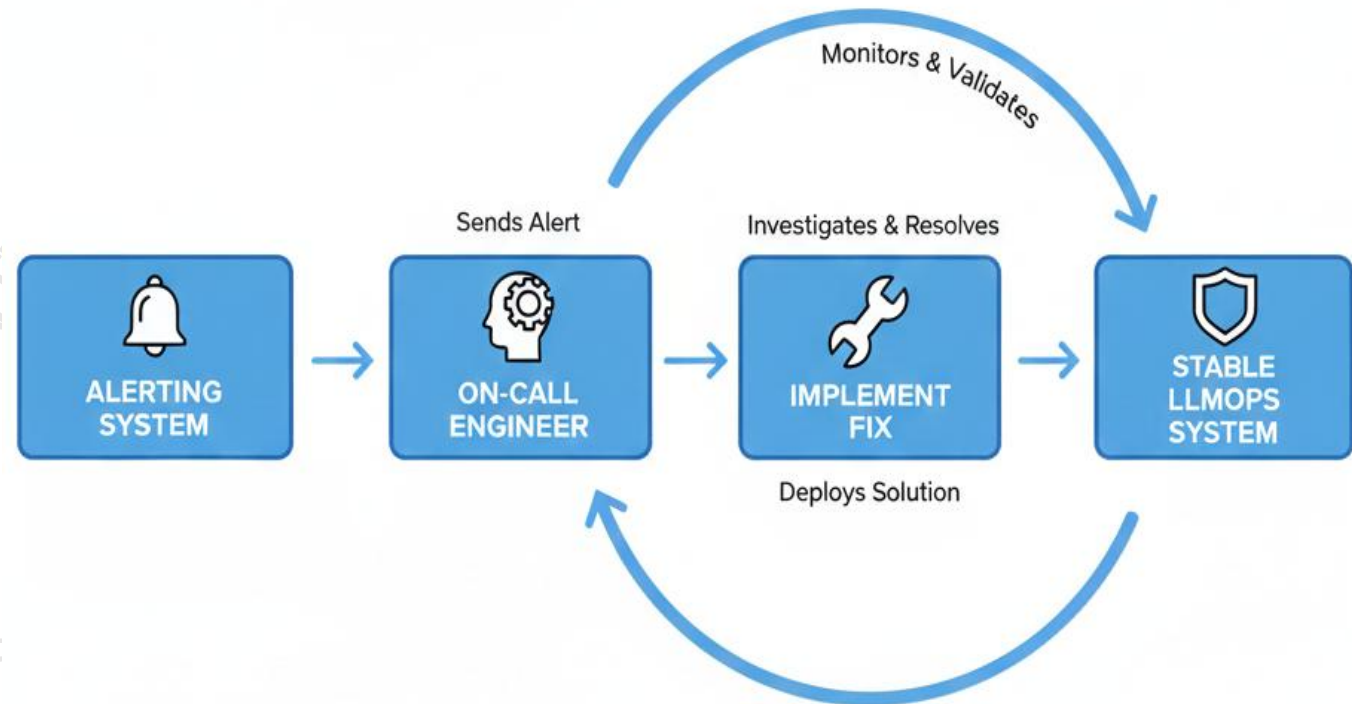
Alerting, Incident Handling & Observability Tools

When Alerts Trigger

- Latency crosses threshold
- Daily cost limit exceeded
- Error rate spikes
- LLM API failures

Incident Response Flow

Alert → Investigation → Fix → Deploy → Monitor



Real-World Example

An alert triggers when token usage exceeds budget → traffic throttled automatically.

Alerting, Incident Handling & Observability Tools

Category	What It Does	Why It Matters for LLMOps	Examples Explained
Monitoring	Continuously tracks system health using metrics like CPU, memory, latency, and request count	LLM apps are expensive and slow if not monitored; monitoring helps detect performance degradation and scaling issues early	CloudWatch monitors AWS resources (EC2, ECS, EKS). Prometheus collects and stores time-series metrics for APIs and services
Logging	Records detailed logs of application events, errors, prompts, and responses	Logs help debug failed LLM responses, API errors, prompt issues, and unexpected behavior	ELK Stack (Elasticsearch, Logstash, Kibana) enables centralized log search and visualization. CloudWatch Logs stores and analyzes application logs in AWS
Tracing	Tracks a single request as it flows through multiple services	LLM systems involve many steps (API → RAG → LLM → DB). Tracing helps identify where latency or failures occur	OpenTelemetry provides distributed tracing across microservices and integrates with many monitoring backends

Thank You

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